0572. Subtree of Another Tree

Easy | Category: Trees | LeetCode: https://leetcode.com/problems/subtree-of-an

Problem

- Return true if subRoot appears in root as a node together with all its descendants.

Intuition

- Try every node of root as a candidate and check Same Tree (problem 100).

Brute Force

- That direct approach is O(m n) and is the expected answer.
- Faster: serialize both with null markers and use string matching (KMP) for O(m + n).

Optimal Approach

- If root is null return false.
- If sameTree(root, subRoot) return true.
- Return isSubtree(root.left, subRoot) || isSubtree(root.right, subRoot).

Worked Example

```
root = [3,4,5,1,2], subRoot = [4,1,2].

at 3: same(3..., 4...) false
at 4: same([4,1,2],[4,1,2]) true -> answer true
```

Java Solution

```java
public class SubtreeOfAnotherTree {
    public boolean isSubtree(TreeNode root, TreeNode sub) {
        if (root == null) return sub == null;
        if (same(root, sub)) return true;
        return isSubtree(root.left, sub) || isSubtree(root.right, sub);
    }

    private boolean same(TreeNode a, TreeNode b) {
        if (a == null || b == null) return a == b;
        return a.val == b.val && same(a.left, b.left) && same(a.right, b.right);
    }
}
```

Complexity

- Time: O(m n) worst case. Space: O(h).

Pitfalls

- A matching subtree must match all the way to the leaves: [3,4,5,1,2,null,null,null,null,0] vs [4,1,2] is false.